

CAPACITOR

SUB TOPICS

⇒ Capacitor:

$$C = \frac{M^{-1} L^{-2} T^4}{A^2}$$

1. Dimensional formula

2. SI units : Microfarad

3. Ratio of Charge to PV $C = \frac{q}{V}$

⇒ Capacitors in Series and Parallel connection.

⇒ Energy Stored by Capacitor.

⇒ Redistribution of Charges

⇒ Energy density

⇒ Work done by battery to charge a Capacitor

⇒ R.C Circuits.

CONCEPTS:

$$\psi = \frac{1}{2}(\epsilon_0)(E)^2 \quad \left(E = \frac{a}{\epsilon_0}\right)$$

$$\psi = \frac{\text{Energy}}{\text{Volume}} = \frac{a^2}{2\epsilon_0} \quad \left(a = \frac{Q}{A}\right)$$

(1) $\vec{E} \rightarrow$ Uniform $U = (\psi)(\text{Volume})$

(2) $\vec{E} \rightarrow$ Non Uniform

$$U = \int_a^b \psi dV$$



Discharging $\rightarrow$ Decrease in energy

Charging $\rightarrow$ Increase in energy

$$U = \frac{Q^2}{2C}$$

RC Circuits: $q = 0.632q_0$ ($\because$ when $R_C = \tau$)

$$* q = q_0(1 - e^{-\frac{t}{R_C}})$$

$$* i = \frac{V}{R} e^{-\frac{t}{R_C}}$$

(3) Force acting on Parallel plate Capacitor

$$F = \frac{q^2}{2A\epsilon_0}$$

Capacitors

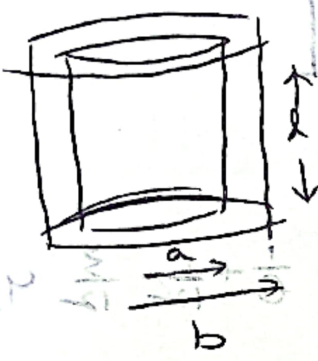
$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$F = \frac{q^2}{2A\epsilon_0}$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$C = \frac{\epsilon_0 A}{d + \frac{t}{k}}$$

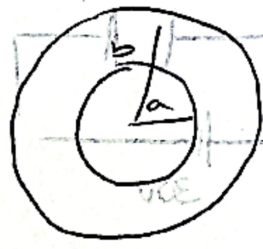
Cylindrical Capacitor:



$$C = \frac{2\pi\epsilon_0 l}{\ln(b/a)}$$

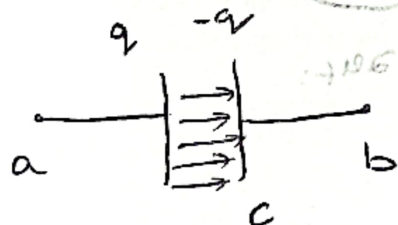
Spherical Capacitor:

$$C = \frac{4\pi\epsilon_0 ab}{b-a}$$



Spherical Conductor:

$$C = 4\pi\epsilon_0 R$$



$$V_a - V_b = V = \frac{q}{C}$$

$$U = \frac{q^2}{2C}$$

$$f = \frac{q^2}{2A\epsilon_0}$$

Energy Density

$$U = \frac{1}{2} \epsilon_0 E^2$$